Supplemental Digital Content 2. Risk of bias assessment – Nonrandomized comparative studies, confounding and selection bias

Study, Year, PMID	1.1 Potential for Any Confounding?	1.2 Potential for Time-Varying Confounding?	1.3 Intervention Switches Related to Prognostic Factors?	1.4 Appropriate Analysis Method for Confounding?	1.5 Appropriate Confounding Variables Used?	1.6 Inappropriate Control of Post-Intervention Variables?	Judgement – Risk of Bias Related to Confounding	2.1 Participant Selection Based on Post-Intervention Variables?	2.2 Post-Intervention Variables Associated with Intervention?	2.3 Post-Intervention Variables Associated with Outcome?	2.4 Start and Follow-Up (Duration) Coincide	2.5 Appropriate Adjustment for Selection Bias	Judgement – Risk of Bias Related to Selection Bias
Abedi, 2016, 25003437	Yes	No	N/A	N	N/A	Y	**Serious**	N	N/A	N/A	Y	N/A	**Low**
Brito, 2020, No PMID	Yes	No	N/A	N	N/A	N	**Serious**	N	N/A	N/A	Y	N/A	**Low**
Carramaschi, 1989, 2602589	Yes	No	N/A	N	N/A	N	**Critical**	N	N/A	N/A	N	N	**Low**
Chetta, 2017, 28002254	Yes	No	N/A	PN	N/A	N	**Serious**	N	N/A	N/A	Y	N/A	**Low**
Dauplat, 2021, 33622886	Yes	No	N/A	Y	Y	Y	**Moderate**	N	N/A	N/A	Y	N/A	**Low**
de Araujo, 2016, 27673527	Yes	No	N/A	PN	N/A	N	**Serious**	N	N/A	N/A	Y	N/A	**Low**
Eltahir, 2015, 25539295	Yes	No	N/A	Y	PY	N	**Low**	N	N/A	N/A	PY	N/A	**Low**
Fischer, 2013, 23629074	Yes	No	N/A	N	N/A	N	**Serious**	N	N/A	N/A	Y	N/A	**Low**
Fischer, 2014, 24916480	Yes	No	N/A	N	N/A	N	**Serious**	N	N/A	N/A	Y	N/A	**Low**
Fischer, 2015, 26366550	Yes	No	N/A	N	N/A	N	**Serious**	N	N/A	N/A	Y	N/A	**Low**
Garbay, 1992, 1624727	Yes	No	N/A	N	N/A	N	**Serious**	N	N/A	N/A	Y	N/A	**Low**
Garvey, 2012, 23096600	Yes	No	N/A	Y	Y	N	**Low**	N	N/A	N/A	Y	N/A	**Low**
Ha, 2020, 32000718	Yes	No	N/A	Y	Y	N	**Low**	N	N/A	N/A	N	PN	**Serious**
Hangge, 2019, 31606126	Yes	No	N/A	PN	N/A	N	**Serious**	N	N/A	N/A	PY	N/A	**Low**
Jiang, 2013, 24349366	Yes	No	N/A	N	N/A	N	**Serious**	N	N/A	N/A	Y	N/A	**Low**
Kouwenberg, 2019, 30270015	Yes	No	N/A	Y	PY	N	**Low**	N	N/A	N/A	PN	PY	**Moderate**
Kouwenberg, 2020, 32590633	Yes	No	N/A	Y	Y	N	**Low**	N	N/A	N/A	PY	N/A	**Low**
Kulkarni, 2017, 28713853	Yes	No	N/A	Y	Y	PN	**Low**	PN	N/A	N/A	Y	N/A	**Low**
Laporta, 2017, 28061518	Yes	No	N/A	PN	N/A	N	**Serious**	N	N/A	N/A	Y	N/A	**Low**
Lei, 2020, 32481367	Yes	No	N/A	Y	Y	Y	**Moderate**	N	N/A	N/A	Y	N/A	**Low**
Liu, 2014, 24558063	Yes	No	N/A	PN	N/A	N	**Serious**	N	N/A	N/A	Y	N/A	**Low**
Mak, 2020, 32665188	Yes	No	N/A	N	N/A	N	**Serious**	N	N/A	N/A	Y	N/A	**Low**
McCarthy, 2014, 24201740	Yes	No	N/A	Y	PY	Y	**Moderate**	PN	N/A	N/A	N	PN	**Moderate**
Merchant, 2015, 26111325	Yes	No	N/A	N	N/A	N	**Serious**	N	N/A	N/A	Y	N/A	**Low**
Mioton, 2013, 23562485	Yes	No	N/A	Y	Y	N	**Low**	PN	N/A	N/A	PY	N/A	**Low**
Momeni, 2018, 29095189	Yes	No	N/A	N	N/A	N	**Serious**	N	N/A	N/A	Y	N/A	**Low**
Naoum, 2020a, 31756414	Yes	No	N/A	PN	N/A	N	**Serious**	N	N/A	N/A	Y	N/A	**Low**
Naoum, 2020b, 32607638	Yes	No	N/A	N	N/A	N	**Serious**	N	N/A	N/A	PY	N/A	**Moderate**

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Nasser, 2018, 30204678	Yes	No	N/A	N	N/A	N	**Serious**	N	N/A	N/A	Y	N/A	**Low**
Nelson, 2019, 31356276	Yes	No	N/A	N	N/A	N	**Serious**	N	N/A	N/A	Y	N/A	**Low**
Palve, 2020, 32468337	Yes	No	N/A	Y	Y	Y	**Moderate**	N	N/A	N/A	Y	N/A	**Low**
Qin, 2018, 29384865	Yes	No	N/A	Y	PY	N	**Low**	N	N/A	N/A	N	PN	**Moderate**
Roth, 2007, 17413877	Yes	No	N/A	N	N/A	N	**Critical**	N	N/A	N/A	Y	N/A	**Low**
Shiraishi, 2020, 32589082	Yes	No	N/A	N	N/A	N	**Serious**	N	N/A	N/A	Y	N/A	**Low**
Simon, 2020, 33363007	Yes	No	N/A	Y	PY	N	**Low**	N	N/A	N/A	Y	N/A	**Low**
Wu, 2021, 33740204	Yes	No	N/A	N	N/A	N	**Serious**	N	N/A	N/A	Y	N/A	**Low**
Woo, 2018, 30360958	Yes	No	N/A	N	N/A	N	**Serious**	N	N/A	N/A	PY	N/A	**Low**
Xu, 2018, 30261115	Yes	No	N/A	Y	PY	N	**Low**	N	N/A	N/A	Y	N/A	**Low**
Yueh, 2009, 19228537	Yes	No	N/A	N	N/A	N	**Critical**	N	N/A	N/A	Y	N/A	**Low**
Zhang, 2019, 30675702	Yes	No	N/A	N	N/A	N	**Serious**	N	N/A	N/A	Y	N/A	**Low**

Abbreviations: N/A = Not applicable, NI = no information, NRCS = nonrandomized comparative study, PMID = PubMed identifier, PN = probably no, PY = probably yes.
Judgements are color coded for emphasis only. The colors do not impart unique information. Signaling questions are not color coded for simplicity and because they are only used to inform the judgements.
Responses to Risk of Bias in Nonrandomized Studies of Interventions (ROBINS-I) signaling questions 1.1 to 1.6 and 2.1 to 2.5 are in regular font. (each item rated as Yes, PY, NI, PN, No, or N/A)
Overall judgements about confounding and selection bias are in **bold font**. Each judgement is rated as **Low**, **Moderate**, **Serious**, **Critical**, or **NI**.